Solutions to MP363 Problem Set 4

①

P.1

This problem deals with the properties of the energy eigenstates of the infinite square well, in particular, the expectation values and uncertainties. So, let's go!

(a) To find $\langle X \rangle$, we obviously need to use

$$X\psi(t,x) = x\psi(t,x)$$

so that

$$\langle X \rangle = \int_{-\infty}^{\infty} \psi^*(t,x)\left(X\psi(t,x)\right) dx$$

$$= \int_{-\infty}^{\infty} \psi^*(t,x)\, x\, \psi(t,x)\, dx$$

$$= \int_0^a \sqrt{\frac{2}{a}}\, e^{iE_N t/\hbar} \sin\left(\frac{N\pi x}{a}\right) \cdot x \cdot \sqrt{\frac{2}{a}}\, e^{-iE_N t/\hbar} \sin\left(\frac{N\pi x}{a}\right) dx$$

$$= \frac{2}{a}\int_0^a x \sin^2\left(\frac{N\pi x}{a}\right) dx$$

$$= \frac{1}{a}\int_0^a x\left[1 - \cos\left(\frac{2N\pi x}{a}\right)\right] dx$$

$$= \frac{1}{a}\left[\frac{1}{2}x^2 - \frac{ax}{2N\pi}\sin\left(\frac{2N\pi x}{a}\right) - \frac{a^2}{4N^2\pi^2}\cos\left(\frac{2N\pi x}{a}\right)\right]_0^a$$

$$= \frac{1}{a}\left|\frac{1}{2}a^2 - \frac{a^2}{2N\pi}\sin(2N\pi) - \frac{a^2}{4N^2\pi^2}\cos(2N\pi) + \frac{a^2}{4N^2\pi^2}\right|$$

$$= \boxed{\frac{a}{2}}$$

since $\sin(2N\pi) = 0$ and $\cos(2N\pi) = 1$ for any integer N. Thus, on average, the particle is in the middle of the box – not a huge surprise.

P acts on wavefunctions as

$$P\psi(t,x) = -i\hbar \frac{\partial \psi(t,x)}{\partial x}$$

so

$$\langle P \rangle = \int_{-\infty}^{\infty} \psi^*(t,x)\, P\psi(t,x)\, dx$$

$$= \int_0^a \sqrt{\frac{2}{a}}\, e^{\frac{iE_N t}{\hbar}} \sin\left(\frac{N\pi x}{a}\right)\left(-i\hbar\sqrt{\frac{2}{a}}\, e^{\frac{iE_N t}{\hbar}} \cdot \frac{N\pi}{a}\cos\left(\frac{N\pi x}{a}\right)\right) dx$$

$$= \frac{-2N\pi\hbar i}{a^2}\int_0^a \sin\left(\frac{N\pi x}{a}\right)\cos\left(\frac{N\pi x}{a}\right) dx$$

(2)

$$= -\frac{N\pi\hbar i}{a^2}\int_0^a \sin\left(\frac{2N\pi x}{a}\right)dx = \frac{i\hbar}{2a}\cos\left(\frac{2N\pi x}{a}\right)\Big|_0^a$$

$$= \frac{i\hbar}{2a}\left[\cos(2N\pi) - 1\right] \boxed{= 0}$$

so, on average, the particle moves to the left just as much as it does to the right, and thus its average momentum is zero.

(b) Since X^2 just multiplies a wavefunction by x^2, we have

$$\langle X^2 \rangle = \int_{-\infty}^{\infty} \psi^*(t,x)\, x^2 \psi(t,x)$$

$$= \frac{2}{a}\int_0^a x^2 \sin^2\left(\frac{N\pi x}{a}\right)dx$$

$$= \frac{1}{a}\int_0^a \left(x^2 - x^2\cos\left(\frac{2N\pi x}{a}\right)\right)dx$$

$$= \frac{1}{a}\left[\frac{x^3}{3} - \frac{ax^2}{2N\pi}\sin\left(\frac{2N\pi x}{a}\right) - \frac{a^2 x}{2N^2\pi^2}\cos\left(\frac{2N\pi x}{a}\right) + \frac{a^3}{4N^3\pi^3}\sin\left(\frac{2N\pi x}{a}\right)\right]_0^a$$

$$= \frac{1}{a}\left[\frac{a^3}{3} + \frac{a^3}{2N^2\pi^2}\cos(2N\pi) + \frac{a^3}{4N^3\pi^3}\sin(2N\pi)\right]$$

$$= \boxed{\frac{a^2}{3} - \frac{a^2}{2N^2\pi^2}}$$

$\langle P^2 \rangle$ is easy to compute, because

$$P^2\psi = \left(-i\hbar\frac{\partial}{\partial x}\right)^2\psi = -\hbar^2\frac{\partial^2\psi}{\partial x^2}$$

$$= -\hbar^2\frac{\partial^2}{\partial x^2}\left(\sqrt{\frac{2}{a}}\, e^{-\frac{iE_N t}{\hbar}}\sin\left(\frac{N\pi x}{a}\right)\right)$$

$$= -\hbar^2\left(\sqrt{\frac{2}{a}}\, e^{-\frac{iE_N t}{\hbar}} \cdot -\frac{N^2\pi^2}{a^2}\sin\left(\frac{N\pi x}{a}\right)\right)$$

$$= \frac{N^2\pi^2\hbar^2}{a^2}\psi$$

Thus,

$$\langle P^2 \rangle = \int_{-\infty}^{\infty}\psi^*(P^2\psi)\,dx$$

$$= \int_{-\infty}^{\infty}\psi^*\left(\frac{N^2\pi^2\hbar^2}{a^2}\psi\right)dx$$

$$= \frac{N^2\pi^2\hbar^2}{a^2}\int_{-\infty}^{\infty}|\psi|^2 dx = \boxed{\frac{N^2\pi^2\hbar^2}{a^2}}$$

because $\int_{-\infty}^{\infty} |\psi|^2 dx = 1$.

(c) We can now compute the uncertainties:

$$\Delta x^2 = \langle X^2 \rangle - \langle X \rangle^2$$

$$= \frac{a^2}{3} - \frac{a^2}{2N^2\pi^2} - \frac{a^2}{4}$$

$$= \frac{a^2}{12} - \frac{a^2}{2N^2\pi^2}$$

or

$$\boxed{\Delta x = \frac{a}{2}\sqrt{\frac{1}{3} - \frac{2}{N^2\pi^2}}}$$

$\langle P \rangle = 0$, so

$$\Delta p = \sqrt{\langle P^2 \rangle} = \boxed{\frac{N\pi\hbar}{a}}$$

and thus

$$\Delta x \Delta p = \frac{a}{2}\sqrt{\frac{1}{3} - \frac{2}{N^2\pi^2}} \cdot \frac{N\pi\hbar}{a}$$

$$= \frac{\hbar}{2}\sqrt{(N\pi)^2\left(\frac{1}{3} - \frac{2}{N^2\pi^2}\right)}$$

$$= \frac{\hbar}{2}\sqrt{\frac{\pi^2 N^2}{3} - 2}$$

I write it this way because it's easy to compare this with $\hbar/2$: note that (i) $N \geq 1$; (ii) $\pi/3 > 1$; and (iii) $\pi > 3$. Thus $\frac{\pi}{3} \cdot \pi \cdot N^2 > 1 \cdot 3 \cdot 1$, or $\frac{\pi^2 N^2}{3} > 3$. Thus, $\frac{\pi^2 N^2}{3} - 2 > 1$, and thus $\sqrt{\frac{\pi^2 N^2}{3} - 2} > 1$. This thus confirms that $\boxed{\Delta x \Delta p > \hbar/2}$, as it must be. (In fact, the ground state wavefunction with $N = 1$ has the least value of $\Delta x \Delta p$, $\frac{\hbar}{2}\sqrt{\pi^2/3 - 2} \approx 1.136 \cdot \hbar/2$.)

P. 2

This problem looks at a particle in a box that isn't in an energy eigenstate, namely,

$$\psi(x) = \begin{cases} 0 & x \leq 0,\ x \geq a \\ Cx(a-x) & 0 < x < a. \end{cases}$$

(Note that $\psi(0) = \psi(a) = 0$, which is required by continuity of ψ. Thus, this is indeed a possible state of the system.)

4

(a) Obviously, $\int_{-\infty}^{\infty}|\psi(x)|^2dx = 1$, so since $\psi = 0$ outside the box,

$$\int_{-\infty}^{\infty}|\psi(t,x)|^2dx = \int_0^a C^2x^2(a-x)^2dx = C^2\int_0^a(a^2x^2-2ax^3+x^4)dx$$

$$= C^2\left(\frac{a^2x^3}{3}-\frac{ax^4}{2}+\frac{x^5}{5}\right|_0^a = C^2\left(\frac{a^5}{3}-\frac{a^5}{2}+\frac{a^5}{5}\right)$$

$$= \frac{C^2a^5}{30}$$

so, assuming C is real and positive, $\boxed{C = \sqrt{\frac{30}{a^5}}}$.

(b) Computing all these expectation values is easy, because ψ and all of its derivatives are simply polynomials, whose integrals are piss-easy. So let's go:

$$\langle X\rangle = \int_{-\infty}^{\infty}\psi^*(x)\,X\,\psi(x)\,dx$$

$$= \int_0^a \sqrt{\frac{30}{a^5}}\,x(a-x)\cdot x\cdot\sqrt{\frac{30}{a^5}}\,x(a-x)\,dx$$

$$= \frac{30}{a^5}\int_0^a x^3(a-x)^2dx.$$

Now, I could write this out and compute it, but notice symmetry: let $y = \frac{a}{2}-x$ be a new variable of integration. Then

$$\langle X\rangle = \frac{30}{a^5}\int_{a/2}^{-a/2}\left(\frac{a}{2}-y\right)^3\left(\frac{a}{2}+y\right)^2(-dy)$$

$$= \frac{30}{a^5}\int_{-a/2}^{a/2}\left(\frac{a^2}{4}-y^2\right)^2\left(\frac{a}{2}-y\right)dy$$

$$= \frac{30}{a^5}\int_{-a/2}^{a/2}\frac{a}{2}\left(\frac{a^2}{4}-y^2\right)^2dy - \frac{30}{a^5}\int_{-a/2}^{a/2}y\left(\frac{a^2}{4}-y^2\right)^2dy.$$

Notice that the first integrand is an even function of y and the second is an odd function. Since the integrals are from $-a/2$ to $a/2$, the second integral vanishes, and the first is just twice the integral from 0 to $a/2$, so

$$\langle X\rangle = \frac{30}{a^5}\cdot 2\int_0^{a/2}\frac{a}{2}\left(\frac{a^2}{4}-y^2\right)^2dy = \frac{30}{a^4}\int_0^{a/2}\left(\frac{a^4}{16}-\frac{a^2y^2}{2}+y^4\right)dy$$

$$= \frac{30}{a^4}\left(\frac{a^4}{16}y-\frac{a^2y^3}{6}+\frac{y^5}{5}\right|_0^{a/2} = \boxed{\frac{a}{2}}$$

which also makes sense: this wavefunction is symmetric around $x = a/2$, and all such wavefunctions will have $\langle X\rangle = a/2$.

$\langle X^2 \rangle$, however, is a bit less intuitive:

$$\langle X^2 \rangle = \int_{-\infty}^{\infty} \psi^*(x) X^2 \psi(x) dx = \frac{30}{a^5} \int_0^a x^4 (a-x)^2 dx$$

$$= \frac{30}{a^5} \int_0^a (a^2x^4 - 2ax^5 + x^6) dx$$

$$= \frac{30}{a^5} \left(\frac{a^2x^5}{5} - \frac{ax^6}{3} + \frac{x^7}{7} \right)\Big|_0^a = \frac{30}{a^5} \left(\frac{a^7}{5} - \frac{a^7}{3} + \frac{a^7}{7} \right)$$

$$= \boxed{\frac{2a^2}{7}}$$

Since $P\psi = -i\hbar \frac{\partial}{\partial x} \left[\sqrt{\frac{30}{a^5}} x(a-x) \right] = -i\hbar \sqrt{\frac{30}{a^5}} (a-2x)$, we have

$$\langle P \rangle = \int_{-\infty}^{\infty} \psi^*(x) P\psi(x) dx = -\frac{30 i\hbar}{a^5} \int_0^a x(a-2x)(a-x) dx$$

$$= -\frac{30 i\hbar}{a^5} \int_0^a (a^2x - 3ax^2 + 2x^3) dx$$

$$= -\frac{30 i\hbar}{a^5} \left(\frac{a^2x^2}{2} - ax^3 + \frac{x^4}{2} \right)\Big|_0^a = -\frac{30 i\hbar}{a^5} \left(\frac{a^4}{2} - a^4 + \frac{a^4}{2} \right)$$

$$\boxed{= 0}$$

$$P^2\psi = P(P\psi) = -i\hbar \frac{\partial}{\partial x} \left(-i\hbar \sqrt{\frac{30}{a^5}} (a-2x) \right) = 2\hbar^2 \sqrt{\frac{30}{a^5}}$$

so

$$\langle P^2 \rangle = \int_0^a \left(\sqrt{\frac{30}{a^5}} x(a-x) \right) \left(2\hbar^2 \sqrt{\frac{30}{a^5}} \right) dx$$

$$= \frac{60\hbar^2}{a^5} \int_0^a (ax - x^2) dx = \frac{60\hbar^2}{a^5} \left(\frac{ax^2}{2} - \frac{x^3}{3} \right)\Big|_0^a = \boxed{\frac{10\hbar^2}{a^2}}$$

Now, we already computed $\langle P^2 \rangle$, but $H = \frac{P^2}{2m}$ here, so $\langle H \rangle = \frac{\langle P^2 \rangle}{2m}$, giving

$$\boxed{\langle H \rangle = \frac{5\hbar^2}{ma^2}}$$

(c) $\Delta x = \sqrt{2a^2/7 - (a/2)^2} = \frac{a}{\sqrt{28}}$ and $\Delta p = \sqrt{10\hbar^2/a^2 - 0} = \frac{\hbar}{a}\sqrt{10}$,

so

$\Delta x \Delta p = \hbar \sqrt{\frac{10}{28}} = \frac{\hbar}{2} \sqrt{\frac{10}{7}}$, and since $\sqrt{10/7} \approx 1.195$, we see that $\boxed{\Delta x \Delta p \geq \hbar/2}$ is indeed satisfied.

(d) To determine the probability that we get E_2, we must know the coefficient α_2 in the expansion

$$\psi(x) = \sum_{n=1}^{\infty} \alpha_n u_n(x) = \sqrt{\frac{2}{a}} \sum_{n=1}^{\infty} \alpha_n \sin\left(\frac{n\pi x}{a}\right).$$

We know that we'll obtain this by taking the inner product of ψ with u_n: $\alpha_n = \langle u_n|\psi\rangle$. Although we only want α_2, let's find all the coefficients:

$$\alpha_n = \langle u_n|\psi\rangle = \int_{-\infty}^{\infty} u_n^*(x)\,\psi(x)\,dx$$

$$= \frac{\sqrt{60}}{a^3} \int_0^a \sin\left(\frac{n\pi x}{a}\right) x(a-x)\,dx$$

$$= \frac{\sqrt{60}}{a^3}\left[\int_0^a ax\sin\left(\frac{n\pi x}{a}\right)dx - \int_0^a x^2 \sin\left(\frac{n\pi x}{a}\right)dx\right]$$

$$= \frac{\sqrt{60}}{a^3}\Big[-\frac{a^2x}{n\pi}\cos\left(\frac{n\pi x}{a}\right) + \frac{a^3}{n^2\pi^2}\sin\left(\frac{n\pi x}{a}\right) + \frac{ax^2}{n\pi}\cos\left(\frac{n\pi x}{a}\right) - \frac{2a^2x}{n^2\pi^2}\sin\left(\frac{n\pi x}{a}\right) - \frac{2a^3}{n^3\pi^3}\cos\left(\frac{n\pi x}{a}\right)\Big|_0^a$$

$$= \frac{\sqrt{60}}{a^3}\Big[-\frac{a^3}{n\pi}\cos(n\pi) + \frac{a^3}{n^2\pi^2}\sin(n\pi) + \frac{a^3}{n\pi}\cos(n\pi) - \frac{2a^3}{n^2\pi^2}\sin(n\pi) - \frac{2a^3}{n^3\pi^3}\cos(n\pi) + \frac{2a^3}{n^3\pi^3}\Big]$$

$\cos(n\pi) = (-1)^n$ and $\sin(n\pi) = 0$, so

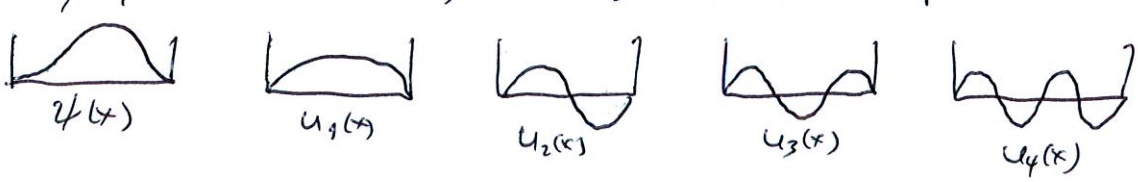
$$\alpha_n = \frac{2\sqrt{60}\,[1-(-1)^n]}{n^3\pi^3}.$$

Note if n is odd, $(-1)^n = -1$ and if n is even, $(-1)^n = 1$, so

$$\alpha_n = \begin{cases} \frac{4\sqrt{60}}{n^3\pi^3} & n = 1,3,5,\dots \\ 0 & n = 2,4,6,\dots \end{cases}$$

So the probability of getting an energy E_2 is $\boxed{\text{zero.}}$

This makes sense: $\psi(x)$ is symmetric about $x = a/2$, as are all the odd eigenstates. However, all the even eigenstates are antisymmetric. Thus, ψ cannot have any even eigenstates in its expansion.



Thus, this argument also gives $\alpha_n = 0$ if n is even.